Assignment-I

Programming Assignment: Experimental Analysis of YOLO Object Detection

Objective

Implement a **YOLO-based object detection system** and investigate how changes in image quality affect its detection performance.

Problem Statement

Select a real-world image or a short video containing multiple objects and use a pretrained YOLO model to detect them. Then, systematically modify the same input and analyze how these changes affect the detector's predictions.

Tasks

1. YOLO Detection

- Select one image or short video containing at least **3 different object instances**.
- Apply a pretrained YOLO model for object detection.
- Display the detected objects with:
    - Bounding boxes
    - Class labels
    - Confidence scores

2. Create Modified Inputs

Create three modified versions of the same input:

- Reduced brightness
- Blurred image
- Reduced brightness + blurred image

Run the **same YOLO model with the same settings** on all four inputs.

3. Performance Analysis

Record the following for each condition:

Condition	Objects Detected	Average Confidence	Inference Time	Missed Objects

Original	—	—	—	—
Reduced Brightness	—	—	—	—
Blurred	—	—	—	—
Brightness + Blur	—	—	—	—

4. Object-Level Investigation

Select **two objects** from the original image and track their detection results across all four conditions.

For each object, report:

- Whether it was detected
- Confidence score
- Bounding-box changes
- Whether the object was missed under any condition

Calculate:

$$\Delta C = C_{original} - C_{modified}$$

5. Critical Analysis

Identify **one interesting change or failure case** in the YOLO predictions.

For example:

- An object was detected in the original image but missed after modification.
- The confidence score decreased substantially.
- The bounding box shifted or became inaccurate.
- A different object was detected instead.

Explain **why you think the detection changed**, using evidence from your actual images.

Your explanation should relate the observation to factors such as:

- Object boundaries

- Texture
- Contrast
- Blur
- Background similarity
- Visibility of discriminative features

6. Required Submission

1. Original image/video
2. Three modified inputs
3. YOLO detection results for all four conditions
4. Completed comparison table
5. Confidence-change calculations
6. One detailed failure-case analysis
7. Short observation and conclusion

7. Important Experimental Conditions

Use:

- The **same input**
- The **same YOLO model**
- The **same confidence threshold**
- The **same experimental procedure**

for all four conditions.

> **The objective is not merely to run YOLO. You must investigate, compare, and explain how image conditions influence object detection.**

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Assignment–2 (CO4)
U-Net for Semantic Image Segmentation

Objective

Study and explain the U-Net architecture and its role in semantic image segmentation, with emphasis on its encoder–decoder structure, skip connections, and pixel-level prediction.

Task
Prepare a PowerPoint presentation on U-Net for Semantic Image Segmentation.
The presentation should cover:

1. Introduction to Semantic Segmentation
- What is semantic segmentation?
- How is it different from image classification?
- How is it different from object detection?
- How is it different from instance segmentation?

2. Introduction to U-Net
- What is U-Net?
- Why was U-Net proposed?
- What type of image-processing problem does it address?

3. U-Net Architecture
- Encoder / Contracting path
- Bottleneck
- Decoder / Expanding path
- Convolutional layers
- Downsampling
- Upsampling
- Skip connections

4. Explain the Architecture with a Diagram
- Create or annotate a U-Net architecture diagram.
- Clearly identify the encoder, bottleneck, decoder, and skip connections.
- Explain the purpose of each major component.

5. Working of U-Net
- Explain the complete flow from input image to segmentation map.
- Explain feature extraction and downsampling.
- Explain the bottleneck.
- Explain upsampling and recovery of spatial information.
- Explain how the final segmentation map is produced.

6. Importance of Skip Connections
- What information is lost during downsampling?
- Why is this information important for segmentation?
- How do skip connections help recover fine spatial details?
- Why are skip connections useful for accurately segmenting object boundaries?

7. Output of U-Net
- Explain how U-Net produces a pixel-wise segmentation output.
- Include an example showing Input Image → Ground Truth/Mask → Predicted Segmentation.

8. Applications
- Discuss at least three applications of U-Net in image processing.
- For each application, briefly explain why pixel-level segmentation is useful.

9. Advantages and Limitations
- Discuss the major advantages of U-Net.
- Discuss its limitations and challenges.

10. Comparison
- Compare U-Net with at least one other image segmentation approach.
- Compare architecture, output, pixel-level segmentation, training requirements, computational complexity, and applications.

11. Case Study
- Select one real-world image segmentation problem and explain how U-Net could be used to solve it.
- Include the input, expected segmentation classes, segmentation mask, expected output, and why U-Net is suitable.

12. References
- Include the original U-Net research paper and other reliable references used for preparing the presentation.

Presentation Requirements
- Prepare a well-structured PPT.
- Use appropriate diagrams, illustrations, and examples.
- Do not fill slides with copied text.
- Explain the architecture using your own understanding.
- Clearly label all figures and diagrams.
- Include references for externally sourced figures or information.

Suggested Presentation Structure
Introduction → Architecture → Working → Skip Connections → Output → Applications → Comparison → Case Study

Submission
Submit the final PPT under: CO4 → Assignment–2

Important
The objective is not simply to describe U-Net. The presentation should demonstrate that you understand how the encoder, bottleneck, decoder, and skip connections work together to produce pixel-level segmentation.